



CUSTOM AI FOR FINTECH

The reference architecture for regulated workloads.

Swipe →





THE ARCHITECTURE PROBLEM



Demos work. Production doesn't.

- Public endpoints retain data beyond policy
- No SOC 2 or SOX-compliant audit trail
- OpenAI Assistants API not designed for FINRA
- Demo success rate does not predict prod reliability





CORE ARCHITECTURE LAYERS



Four layers. One compliant stack.

- Private inference on VPC-isolated infrastructure
- Agentic orchestration owned by the client team
- Compliance-first audit log as foundational layer
- Human reviewer queue as first-class deliverable





MODEL SELECTION



Right model for the risk level.

- High-risk decisions: private fine-tuned model
- Medium-risk: managed API with data redaction
- Low-risk: commercial API on non-sensitive data
- Latency and cost model by workload tier





ORCHESTRATION PATTERNS



Orchestration that holds under load.

- State management designed for multi-turn compliance
- Tool use scoped to approved data sources only
- Retry logic with audit log on every failure
- Graceful degradation: agent falls back to human queue





INTEGRATION REQUIREMENTS



Connecting agents to existing systems.

- Core banking and CRM read-only by default
- Write access gated by compliance approval workflow
- Webhook output to downstream compliance systems
- API versioning tied to regulatory change schedule





SECURITY REQUIREMENTS



Security for regulated AI ops.

- VPC deployment with no model data egress
- Encryption at rest and in transit for all agent memory
- Role-based access to reviewer queue and audit log
- Penetration test and third-party audit before go-live





TEAM SHAPE



Who you need to ship in prod.

- ML engineer for model integration and fine-tuning
- Compliance engineer for audit trail and controls design
- Backend engineer for orchestration and API integration
- Compliance team as stakeholder from week one





CODISTE AI

Ready to build your regulated agent stack?

Talk to us ↗

codiste.com